Sagar Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.92
Holy mother sharda sr. sec. school Kota, rajasthan	Class XII (RBSE)	2024	93.40%
Govt. sr. sec. school bhinmal	Class X (RBSE)	2022	93.17%

SCHOLASTIC ACHIEVEMENTS

Solved 300+ LeetCode and 230+ Codeforces problems in data structures and algorithms.

Recipient of the **Anuprati Scholarship**, Government of Rajasthan (2022).

Codeforces Max Rating: 1370.

SKILLS

- **Programming Languages:** C++, C, Python, SQL, JavaScript
- **Libraries:** NumPy, Pandas, Scikit-learn, SDL2, libpcap, Matplotlib
- **Tools & Platforms:** Git, Linux

PROJECTS

Network Packet Sniffer | C++, Libpcap, Linux (Link)

- Developed a packet sniffer in C++ using libpcap for live packet capture and protocol analysis.
- Implemented packet parsing for **Ethernet, IPv4, TCP, UDP, and ICMP** protocols with detailed header extraction and protocol statistics.
- Implemented protocol filtering, packet inspection, and customizable traffic analysis for network debugging and monitoring.

PDF Malware Detection System | (Link)

- Developed an end-to-end malware detection pipeline combining static file analysis with **Random Forest** to classify malicious vs. benign PDFs.
- Engineered a **custom Python structural parser** to extract complex metadata features (JavaScript objects, stream counts, cross-reference tables).
- Evaluated multiple algorithms (**SVM, Stacking, Random Forest**) on the **CIC-Evasive-PDFMal2022** dataset to maximize classification accuracy.

Substring Matching Using Suffix Trees | C++ (Link)

- Done under the guidance of Prof. J. Balasubramaniam as part of the Data Structures and Applications Laboratory course.
- Implemented **Ukkonen's algorithm** for linear-time **O(n)** suffix tree construction.
- Built a high-performance substring search engine supporting **O(m + |pattern|)** query time.
- Benchmarked the implementation against a naive substring search on a **10,000-character** dataset with short query patterns, achieving **10×-20×** faster search performance.

Applied Statistics Coursework | **Python, NumPy, Pandas, Matplotlib**

- Conducted exploratory data analysis through frequency tables, histograms, bar charts, pie charts, and box plots to summarize data distributions and identify outliers.
- Applied point estimation, confidence intervals, and sampling distributions to estimate population parameters and interpret statistical results.

- Performed hypothesis testing, contingency table analysis, and linear regression to evaluate relationships between variables and draw data-driven conclusions.

RELEVANT COURSES

- **Computer Science:** Data Structures & Applications, Algorithms, Theory of Computation, Database Management Systems, Compiler Design, Introduction to Programming
- **Mathematics:** Linear Algebra, Probability Theory, Applied Statistics, Convex Optimization, Number Theory, Introduction to Group Theory, Ordinary Differential Equations, Complex Variables

POSITIONS OF RESPONSIBILITY

Entrepreneurship Cell (E-Cell), IIT Hyderabad

Operations Manager (2025-26)

Assistant Operations Manager (2024-25)

- Managed operations for flagship events E-Summit and Emerge, coordinating logistics and collaborating with multiple teams for seamless event execution.

National Service Scheme (NSS), IIT Hyderabad

Volunteer

- Participated in institute and community service initiatives.